

Unit 3 Pre-Test

Solving Linear Equations

Standard assessed: NC.8.EE.7

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.EE.7 Solve linear equations in one variable.

• I can solve one-step and two-step equations using inverse operations.

Match each vocabulary term to its definition, then solve each equation.

C ▾ coefficient ✓	A. Terms with the same variable raised to the same power.
E ▾ constant ✓	B. The value of the variable that makes the equation true.
D ▾ inverse operations ✓	C. A number multiplied by a variable.
B ▾ solution ✓	D. Operations that undo each other, such as + and −.
F ▾ distributive property ✓	E. A term with no variable — just a number.
A ▾ like terms ✓	F. Multiplying a factor across each term inside parentheses.

Keep the equation balanced — worked example

$$2x + 5 = 13 \rightarrow \text{subtract } 5 \rightarrow 2x = 8 \rightarrow \text{divide by } 2 \rightarrow \mathbf{x = 4}$$

Whatever you do to one side, do to the other.

1. Solve

$$x + 7 = 12$$

x =

2. Solve

$$x - 4 = 9$$

x =

answer: $x = 5$

3. Solve

$$3x = 21$$

$x =$

answer: $x = 7$

5. Solve

$$2x + 3 = 11$$

$x =$

answer: $x = 4$

answer: $x = 13$

4. Solve

$$x \div 5 = 6$$

$x =$

answer: $x = 30$

6. Solve

$$4x + 5 = 25$$

$x =$

answer: $x = 5$

SCORE 4.0 Target Content

NC.8.EE.7 Solve equations with the distributive property and variables on both sides.

• *I can solve multi-step equations, including variables on both sides.*

Solve each equation. Show your work.

7. Solve

$$2(x + 3) = 14$$

$x =$

answer: $x = 4$

9. Solve

$$5x + 2 = 3x + 10$$

$x =$

answer: $x = 4$

11. Solve

8. Solve

$$3(x - 2) = 9$$

$x =$

answer: $x = 5$

10. Solve

$$7x - 4 = 2x + 11$$

$x =$

answer: $x = 3$

12. Solve

$$2(x + 4) = 3x + 5$$

x =

answer: $x = 3$

$$3(2x - 1) = 4x + 7$$

x =

answer: $x = 5$

SCORE **5.0** Advanced Content

NC.8.EE.7 Solve equations with fractions or decimals and identify special solutions.

• *I can solve equations with rational coefficients and recognize equations with one, none, or infinitely many solutions.*

Solve. If there is no solution or infinitely many, write that instead.

13. Solve

$$\frac{1}{2}x + 3 = 9$$

x =

answer: $x = 12$

14. Solve

$$0.2x + 1 = 3$$

x =

answer: $x = 10$

15. Solve

$$\frac{2}{3}x = 8$$

x =

answer: $x = 12$

16. Solve

$$3x + 5 = 3x + 5$$

Answer:

answer: infinitely many solutions

17. Solve

$$2x + 4 = 2x - 1$$

Answer:

answer: no solution

18. Solve

$$5(x + 2) - 3 = 2(x + 5)$$

x =

answer: $x = 1$